

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',          # L1 con penalty annealing
    'capa': 6,                      # Layer del modelo OthelloGPT
    'juegos': 1000,                 # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments,
    'extras': {
        'expansion': 32,
        'sparsity_coef': 1e-2,
        'epochs': 250,
        'patience': 20,
        'learning_rate': 3e-4,
        'sparsity_decay': 0.98,
```

```
Metadata de naming:
  variante: standard
  tecnica: l1-decay
  capa: 6
  juegos: 1000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 32, 'sparsity_coef': 0.01, 'epochs': 250, 'patience': 20, 'learning_
```

Nuevo extras registrado: ex4 → {'expansion': 32, 'sparsity_coef': 0.01, 'epochs': 250, 'pat
extras_id: ex4

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```

# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 256,
    'num_epochs': 250,
    'learning_rate': 3e-4,
    'sparsity_coef': 1e-2,
    'sparsity_decay': 0.98,

    # Early Stopping
    'early_stopping': True,
    'patience': 20,

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['juegos']}.npy"),
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

```

```

# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

```

=====
CONFIGURACIÓN DEL SAE
=====

```

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	16384
Arquitectura	tied_weights	False
Entrenamiento	batch_size	256
Entrenamiento	num_epochs	250
Entrenamiento	learning_rate	0.0003

```

Entrenamiento sparsity_coef 0.01
Entrenamiento sparsity_decay 0.98
Early Stopping early_stopping True
Early Stopping patience 20
=====

Archivo de activaciones: ..\..\..\activations\data\layer6_1000games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex4\plots
=====

```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})

print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')

```

```

print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

```

```

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====

```

	Propiedad	Valor
	Total muestras	59,000
	Shape	(59000, 512)
	Dtype	float32
Tamaño en memoria		115.23 MB

```

=====

=====
SPLIT DE DATOS (80-20)
=====

```

	Conjunto	Muestras	Porcentaje
	Train	47,200	80.0%
Validation		11,800	20.0%
	Total	59,000	100.0%

```

=====

```


=====				
DATALOADERS				
=====				
Loader	Batches	Batch Size	Shuffle	
Train	185	256	Sí	
Validation	47	256	No	
=====				

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']}  $\rightarrow$  {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']}  $\rightarrow$  {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '_'
    ],
    'Parámetros': [
```

```

        '-',
        f"{encoder_params:},{",
        '-',
        f"{decoder_params:},{",
        '-',
        f"{num_params:},{",
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO
=====

```

	Componente	Dimensión	Parámetros
	Input	512	-
	Encoder	512 → 16384	8,404,992
Hidden	(Latent)	16384	-
	Decoder	16384 → 512	8,389,120
	Output	512	-
	Total	-	16,794,112

```

=====

Factor de expansión: 32.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE:** Mide qué tan bien reconstruimos las activaciones originales
- **L1:** Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay'],
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
=====
OPTIMIZACIÓN
=====
                Parámetro Valor
            Optimizador  Adam
            Learning Rate 3e-04
Sparsity Coef (inicial) 1e-02
            Sparsity Decay 0.98
=====
```

7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': [] # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```

# ===== L1 DECAY =====
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ===== ENTRENAMIENTO =====
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={current_sparsity_coef}')

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()

```

```

epoch_mse_loss += mse_loss.item()
epoch_sparsity_loss += sparsity_loss.item()
epoch_l0 += l0
epoch_fraction_active += fraction_active

# Actualizar barra
pbar.set_postfix({
    'loss': f'{total_loss.item():.4f}',
    'mse': f'{mse_loss.item():.4f}',
    'l0': f'{l0:.1f}'
})

# Promedios de entrenamiento
num_train_batches = len(train_loader)
avg_train_total_loss = epoch_total_loss / num_train_batches
avg_train_mse_loss = epoch_mse_loss / num_train_batches
avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
avg_train_l0 = epoch_l0 / num_train_batches
avg_train_fraction_active = epoch_fraction_active / num_train_batches

# ===== VALIDACIÓN =====
model.eval()

val_total_loss = 0
val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # Calcular pérdida con sparsity_coef actualizado
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, current_sparsity_coef
        )

# Métricas

```

```

10 = (hidden > 0).float().sum(dim=1).mean().item()
fraction_active = (hidden > 0).float().mean().item()

# Acumular
val_total_loss += total_loss.item()
val_mse_loss += mse_loss.item()
val_sparsity_loss += sparsity_loss.item()
val_l0 += l0
val_fraction_active += fraction_active

# Promedios de validación
num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    # Guardar mejor modelo (local + organizado)
    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),

```

```

        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    # Guardado local
    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

    # Guardado organizado
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=CONFIG,
        history=history,
        optimizer=optimizer
    )

    print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= CONFIG['patience']:
        print(f'\n Early stopping activado en época {epoch+1}')
        print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
        break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:}')

```



```

        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0')
        if CONFIG['early_stopping']:
            print(f'    Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {best_val_mse:.4f}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)

```

Iniciando entrenamiento: 250 épocas máximo
 L1 decay: $1.00e-02 \times 0.98^{\text{época}}$
 Early stopping: patience=20 épocas

=====

Época 1/250 [Train] L1=1.00e-02: 100%| | 185/185 [00:02<00:00, 81.37it/s, loss=0.8315, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 1 (Val MSE: 0.6668)

Época 1/250 | L1 coef: 1.00e-02

Train - Total: 1.3941 | MSE: 0.7424 | L0: 518.0

Val - Total: 0.7333 | MSE: 0.6668 | L0: 35.7

Best Val MSE: 0.6668 @ época 1 | Sin mejora: 0/20

Época 2/250 [Train] L1=9.80e-03: 100%| | 185/185 [00:02<00:00, 91.80it/s, loss=0.6733, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 2 (Val MSE: 0.6156)

Época 3/250 [Train] L1=9.60e-03: 100%| | 185/185 [00:02<00:00, 91.89it/s, loss=0.6249, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 3 (Val MSE: 0.5894)

Época 4/250 [Train] L1=9.41e-03: 100%| | 185/185 [00:02<00:00, 91.36it/s, loss=0.6295, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 4 (Val MSE: 0.5721)

Época 5/250 [Train] L1=9.22e-03: 100%| | 185/185 [00:02<00:00, 90.63it/s, loss=0.6082, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 5 (Val MSE: 0.5587)

Época 5/250 | L1 coef: 9.22e-03

Train - Total: 0.6077 | MSE: 0.5643 | L0: 4.3

Val - Total: 0.6046 | MSE: 0.5587 | L0: 12.8

Best Val MSE: 0.5587 @ época 5 | Sin mejora: 0/20

Época 6/250 [Train] L1=9.04e-03: 100%| | 185/185 [00:02<00:00, 90.86it/s, loss=0.5785, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 6 (Val MSE: 0.5478)

Época 7/250 [Train] L1=8.86e-03: 100%| | 185/185 [00:02<00:00, 86.56it/s, loss=0.5892, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 7 (Val MSE: 0.5385)

Época 8/250 [Train] L1=8.68e-03: 100%| | 185/185 [00:02<00:00, 79.96it/s, loss=0.5580, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 8 (Val MSE: 0.5293)

Época 9/250 [Train] L1=8.51e-03: 100%| | 185/185 [00:02<00:00, 81.11it/s, loss=0.5622, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 9 (Val MSE: 0.5204)

Época 10/250 [Train] L1=8.34e-03: 100%| | 185/185 [00:02<00:00, 82.66it/s, loss=0.5764,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 10 (Val MSE: 0.5109)

Época 10/250 | L1 coef: 8.34e-03

Train - Total: 0.5584 | MSE: 0.5143 | L0: 8.9

Val - Total: 0.5585 | MSE: 0.5109 | L0: 18.4

Best Val MSE: 0.5109 @ época 10 | Sin mejora: 0/20

Época 11/250 [Train] L1=8.17e-03: 100%| | 185/185 [00:02<00:00, 76.52it/s, loss=0.5311,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 11 (Val MSE: 0.5014)

Época 12/250 [Train] L1=8.01e-03: 100%| | 185/185 [00:02<00:00, 80.82it/s, loss=0.5382,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 12 (Val MSE: 0.4905)

Época 13/250 [Train] L1=7.85e-03: 100%| | 185/185 [00:02<00:00, 79.88it/s, loss=0.5244,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 13 (Val MSE: 0.4803)

Época 14/250 [Train] L1=7.69e-03: 100%| | 185/185 [00:02<00:00, 82.96it/s, loss=0.5267,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 14 (Val MSE: 0.4704)

Época 15/250 [Train] L1=7.54e-03: 100%| | 185/185 [00:02<00:00, 77.85it/s, loss=0.5208,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 15 (Val MSE: 0.4603)

Época 15/250 | L1 coef: 7.54e-03

Train - Total: 0.5178 | MSE: 0.4639 | L0: 19.3

Val - Total: 0.5180 | MSE: 0.4603 | L0: 29.2

Best Val MSE: 0.4603 @ época 15 | Sin mejora: 0/20

Época 16/250 [Train] L1=7.39e-03: 100%| | 185/185 [00:02<00:00, 82.33it/s, loss=0.4982,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 16 (Val MSE: 0.4513)

Época 17/250 [Train] L1=7.24e-03: 100%| | 185/185 [00:02<00:00, 80.64it/s, loss=0.5180,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 17 (Val MSE: 0.4421)

Época 18/250 [Train] L1=7.09e-03: 100%| | 185/185 [00:02<00:00, 83.11it/s, loss=0.4973,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 18 (Val MSE: 0.4335)

Época 19/250 [Train] L1=6.95e-03: 100%| | 185/185 [00:02<00:00, 81.06it/s, loss=0.4825,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 19 (Val MSE: 0.4253)

Época 20/250 [Train] L1=6.81e-03: 100%| | 185/185 [00:02<00:00, 78.88it/s, loss=0.4793,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 20 (Val MSE: 0.4185)

Época 20/250 | L1 coef: 6.81e-03

Train - Total: 0.4817 | MSE: 0.4203 | L0: 31.2

Val - Total: 0.4826 | MSE: 0.4185 | L0: 40.8

Best Val MSE: 0.4185 @ época 20 | Sin mejora: 0/20

Época 21/250 [Train] L1=6.68e-03: 100%| | 185/185 [00:02<00:00, 78.93it/s, loss=0.4620,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 21 (Val MSE: 0.4120)

Época 22/250 [Train] L1=6.54e-03: 100%| | 185/185 [00:02<00:00, 81.40it/s, loss=0.4613,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 22 (Val MSE: 0.4059)

Época 23/250 [Train] L1=6.41e-03: 100%| | 185/185 [00:02<00:00, 78.95it/s, loss=0.4605,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 23 (Val MSE: 0.4007)

Época 24/250 [Train] L1=6.28e-03: 100%| | 185/185 [00:02<00:00, 81.72it/s, loss=0.4546,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 24 (Val MSE: 0.3965)

Época 25/250 [Train] L1=6.16e-03: 100%| | 185/185 [00:02<00:00, 79.29it/s, loss=0.4534,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 25 (Val MSE: 0.3914)

Época 25/250 | L1 coef: 6.16e-03

Train - Total: 0.4542 | MSE: 0.3925 | L0: 39.9

Val - Total: 0.4554 | MSE: 0.3914 | L0: 49.2

Best Val MSE: 0.3914 @ época 25 | Sin mejora: 0/20

Época 26/250 [Train] L1=6.03e-03: 100%| | 185/185 [00:02<00:00, 78.05it/s, loss=0.4523,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 26 (Val MSE: 0.3867)

Época 27/250 [Train] L1=5.91e-03: 100%| | 185/185 [00:02<00:00, 82.58it/s, loss=0.4407,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 27 (Val MSE: 0.3827)

Época 28/250 [Train] L1=5.80e-03: 100%| | 185/185 [00:02<00:00, 80.91it/s, loss=0.4402,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 28 (Val MSE: 0.3793)

Época 29/250 [Train] L1=5.68e-03: 100%| | 185/185 [00:02<00:00, 78.65it/s, loss=0.4234,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 29 (Val MSE: 0.3765)

Época 30/250 [Train] L1=5.57e-03: 100%| | 185/185 [00:02<00:00, 79.95it/s, loss=0.4453,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 30 (Val MSE: 0.3739)

Época 30/250 | L1 coef: 5.57e-03

Train - Total: 0.4327 | MSE: 0.3737 | L0: 46.3

Val - Total: 0.4346 | MSE: 0.3739 | L0: 55.4

Best Val MSE: 0.3739 @ época 30 | Sin mejora: 0/20

Época 31/250 [Train] L1=5.45e-03: 100%| | 185/185 [00:02<00:00, 80.79it/s, loss=0.4361,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 31 (Val MSE: 0.3716)

Época 32/250 [Train] L1=5.35e-03: 100%| | 185/185 [00:02<00:00, 76.47it/s, loss=0.4219,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 32 (Val MSE: 0.3692)

Época 33/250 [Train] L1=5.24e-03: 100%| | 185/185 [00:02<00:00, 79.33it/s, loss=0.4326,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 33 (Val MSE: 0.3672)

Época 34/250 [Train] L1=5.13e-03: 100%| | 185/185 [00:02<00:00, 80.26it/s, loss=0.4167,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 34 (Val MSE: 0.3647)

Época 35/250 [Train] L1=5.03e-03: 100%| | 185/185 [00:02<00:00, 80.24it/s, loss=0.4079,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 35 (Val MSE: 0.3628)

Época 35/250 | L1 coef: 5.03e-03
Train - Total: 0.4168 | MSE: 0.3621 | L0: 50.7
Val - Total: 0.4189 | MSE: 0.3628 | L0: 59.8
Best Val MSE: 0.3628 @ época 35 | Sin mejora: 0/20

Época 36/250 [Train] L1=4.93e-03: 100%| | 185/185 [00:02<00:00, 81.02it/s, loss=0.4084,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 36 (Val MSE: 0.3607)

Época 37/250 [Train] L1=4.83e-03: 100%| | 185/185 [00:02<00:00, 78.99it/s, loss=0.4128,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 37 (Val MSE: 0.3590)

Época 38/250 [Train] L1=4.74e-03: 100%| | 185/185 [00:02<00:00, 79.34it/s, loss=0.4179,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 38 (Val MSE: 0.3576)

Época 39/250 [Train] L1=4.64e-03: 100%| | 185/185 [00:02<00:00, 90.15it/s, loss=0.4245,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 39 (Val MSE: 0.3560)

Época 40/250 [Train] L1=4.55e-03: 100%| | 185/185 [00:02<00:00, 90.74it/s, loss=0.4056,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 40 (Val MSE: 0.3548)

Época 40/250 | L1 coef: 4.55e-03
Train - Total: 0.4043 | MSE: 0.3541 | L0: 54.1
Val - Total: 0.4064 | MSE: 0.3548 | L0: 63.1
Best Val MSE: 0.3548 @ época 40 | Sin mejora: 0/20

Época 41/250 [Train] L1=4.46e-03: 100%| | 185/185 [00:02<00:00, 91.05it/s, loss=0.4119,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 41 (Val MSE: 0.3537)

Época 42/250 [Train] L1=4.37e-03: 100%| | 185/185 [00:02<00:00, 86.70it/s, loss=0.4060,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 42 (Val MSE: 0.3524)

Época 43/250 [Train] L1=4.28e-03: 100%| | 185/185 [00:02<00:00, 80.49it/s, loss=0.4066,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 43 (Val MSE: 0.3515)

Época 44/250 [Train] L1=4.19e-03: 100%| | 185/185 [00:02<00:00, 80.41it/s, loss=0.4035,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 44 (Val MSE: 0.3503)

Época 45/250 [Train] L1=4.11e-03: 100%| | 185/185 [00:02<00:00, 79.87it/s, loss=0.3961,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 45 (Val MSE: 0.3493)

Época 45/250 | L1 coef: 4.11e-03
Train - Total: 0.3943 | MSE: 0.3486 | L0: 56.7
Val - Total: 0.3964 | MSE: 0.3493 | L0: 65.7
Best Val MSE: 0.3493 @ época 45 | Sin mejora: 0/20

Época 46/250 [Train] L1=4.03e-03: 100%| | 185/185 [00:02<00:00, 82.31it/s, loss=0.3883,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 46 (Val MSE: 0.3487)

Época 47/250 [Train] L1=3.95e-03: 100%| | 185/185 [00:02<00:00, 82.78it/s, loss=0.3967,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 47 (Val MSE: 0.3478)

Época 48/250 [Train] L1=3.87e-03: 100%| | 185/185 [00:02<00:00, 77.47it/s, loss=0.3917,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 48 (Val MSE: 0.3468)

Época 49/250 [Train] L1=3.79e-03: 100%| | 185/185 [00:02<00:00, 77.68it/s, loss=0.3989,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 49 (Val MSE: 0.3461)

Época 50/250 [Train] L1=3.72e-03: 100%| | 185/185 [00:02<00:00, 82.86it/s, loss=0.3732,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 50 (Val MSE: 0.3454)

Época 50/250 | L1 coef: 3.72e-03
Train - Total: 0.3861 | MSE: 0.3445 | L0: 58.7
Val - Total: 0.3883 | MSE: 0.3454 | L0: 67.6
Best Val MSE: 0.3454 @ época 50 | Sin mejora: 0/20

Época 51/250 [Train] L1=3.64e-03: 100%| | 185/185 [00:02<00:00, 81.49it/s, loss=0.3940,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 51 (Val MSE: 0.3446)

Época 52/250 [Train] L1=3.57e-03: 100%| | 185/185 [00:02<00:00, 78.46it/s, loss=0.3736,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 52 (Val MSE: 0.3439)

Época 53/250 [Train] L1=3.50e-03: 100%| | 185/185 [00:02<00:00, 84.03it/s, loss=0.3801,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 53 (Val MSE: 0.3433)

Época 54/250 [Train] L1=3.43e-03: 100%| | 185/185 [00:02<00:00, 81.24it/s, loss=0.3947,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 54 (Val MSE: 0.3425)

Época 55/250 [Train] L1=3.36e-03: 100%| | 185/185 [00:02<00:00, 81.55it/s, loss=0.3711,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 55 (Val MSE: 0.3420)

Época 55/250 | L1 coef: 3.36e-03
Train - Total: 0.3791 | MSE: 0.3412 | L0: 60.5
Val - Total: 0.3810 | MSE: 0.3420 | L0: 69.3
Best Val MSE: 0.3420 @ época 55 | Sin mejora: 0/20

Época 56/250 [Train] L1=3.29e-03: 100%| | 185/185 [00:02<00:00, 81.07it/s, loss=0.3843,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 56 (Val MSE: 0.3415)

Época 57/250 [Train] L1=3.23e-03: 100%| | 185/185 [00:02<00:00, 81.28it/s, loss=0.3755,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 57 (Val MSE: 0.3411)

Época 58/250 [Train] L1=3.16e-03: 100%| | 185/185 [00:02<00:00, 80.58it/s, loss=0.3798,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 58 (Val MSE: 0.3406)

Época 59/250 [Train] L1=3.10e-03: 100%| | 185/185 [00:02<00:00, 79.10it/s, loss=0.3652,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 59 (Val MSE: 0.3401)

Época 60/250 [Train] L1=3.04e-03: 100%| | 185/185 [00:02<00:00, 77.33it/s, loss=0.3823,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 60 (Val MSE: 0.3394)

Época 60/250 | L1 coef: 3.04e-03
Train - Total: 0.3732 | MSE: 0.3387 | L0: 62.0
Val - Total: 0.3750 | MSE: 0.3394 | L0: 70.9
Best Val MSE: 0.3394 @ época 60 | Sin mejora: 0/20

Época 61/250 [Train] L1=2.98e-03: 100%| | 185/185 [00:02<00:00, 81.39it/s, loss=0.3637,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 61 (Val MSE: 0.3392)

Época 62/250 [Train] L1=2.92e-03: 100%| | 185/185 [00:02<00:00, 79.45it/s, loss=0.3707,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 62 (Val MSE: 0.3391)

Época 63/250 [Train] L1=2.86e-03: 100%| | 185/185 [00:02<00:00, 82.23it/s, loss=0.3703,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 63 (Val MSE: 0.3381)

Época 64/250 [Train] L1=2.80e-03: 100%| | 185/185 [00:02<00:00, 81.11it/s, loss=0.3713,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 64 (Val MSE: 0.3377)

Época 65/250 [Train] L1=2.74e-03: 100%| | 185/185 [00:02<00:00, 80.11it/s, loss=0.3601,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 65 (Val MSE: 0.3374)

Época 65/250 | L1 coef: 2.74e-03
Train - Total: 0.3679 | MSE: 0.3365 | L0: 63.3
Val - Total: 0.3697 | MSE: 0.3374 | L0: 72.0
Best Val MSE: 0.3374 @ época 65 | Sin mejora: 0/20

Época 66/250 [Train] L1=2.69e-03: 100%| | 185/185 [00:02<00:00, 81.53it/s, loss=0.3789,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 66 (Val MSE: 0.3373)

Época 67/250 [Train] L1=2.64e-03: 100%| | 185/185 [00:02<00:00, 78.98it/s, loss=0.3499,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 67 (Val MSE: 0.3367)

Época 68/250 [Train] L1=2.58e-03: 100%| | 185/185 [00:02<00:00, 79.39it/s, loss=0.3829,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 68 (Val MSE: 0.3363)

Época 69/250 [Train] L1=2.53e-03: 100%| | 185/185 [00:02<00:00, 78.58it/s, loss=0.3576,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 69 (Val MSE: 0.3361)

Época 70/250 [Train] L1=2.48e-03: 100%| | 185/185 [00:02<00:00, 82.36it/s, loss=0.3707,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 70 (Val MSE: 0.3356)

Época 70/250 | L1 coef: 2.48e-03
Train - Total: 0.3635 | MSE: 0.3348 | L0: 64.4
Val - Total: 0.3651 | MSE: 0.3356 | L0: 73.2
Best Val MSE: 0.3356 @ época 70 | Sin mejora: 0/20

Época 71/250 [Train] L1=2.43e-03: 100%| | 185/185 [00:02<00:00, 81.40it/s, loss=0.3477,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 71 (Val MSE: 0.3352)

Época 72/250 [Train] L1=2.38e-03: 100%| | 185/185 [00:02<00:00, 79.00it/s, loss=0.3603,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 72 (Val MSE: 0.3350)

Época 73/250 [Train] L1=2.33e-03: 100%| | 185/185 [00:02<00:00, 82.11it/s, loss=0.3509,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 73 (Val MSE: 0.3349)

Época 74/250 [Train] L1=2.29e-03: 100%| | 185/185 [00:02<00:00, 81.08it/s, loss=0.3632,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 74 (Val MSE: 0.3343)

Época 75/250 [Train] L1=2.24e-03: 100%| | 185/185 [00:02<00:00, 80.48it/s, loss=0.3642,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 75 (Val MSE: 0.3342)

Época 75/250 | L1 coef: 2.24e-03
Train - Total: 0.3595 | MSE: 0.3333 | L0: 65.4
Val - Total: 0.3612 | MSE: 0.3342 | L0: 74.1
Best Val MSE: 0.3342 @ época 75 | Sin mejora: 0/20

Época 76/250 [Train] L1=2.20e-03: 100%| | 185/185 [00:02<00:00, 80.90it/s, loss=0.3638,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 76 (Val MSE: 0.3338)

Época 77/250 [Train] L1=2.15e-03: 100%| | 185/185 [00:02<00:00, 80.53it/s, loss=0.3494,
Época 78/250 [Train] L1=2.11e-03: 100%| | 185/185 [00:02<00:00, 77.97it/s, loss=0.3633,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 78 (Val MSE: 0.3335)

Época 79/250 [Train] L1=2.07e-03: 100%| | 185/185 [00:02<00:00, 80.17it/s, loss=0.3658,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 79 (Val MSE: 0.3334)

Época 80/250 [Train] L1=2.03e-03: 100%| | 185/185 [00:02<00:00, 82.65it/s, loss=0.3595,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 80 (Val MSE: 0.3328)

Época 80/250 | L1 coef: 2.03e-03
Train - Total: 0.3560 | MSE: 0.3320 | L0: 66.2
Val - Total: 0.3575 | MSE: 0.3328 | L0: 75.2
Best Val MSE: 0.3328 @ época 80 | Sin mejora: 0/20

Época 81/250 [Train] L1=1.99e-03: 100%| | 185/185 [00:02<00:00, 80.89it/s, loss=0.3698,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 81 (Val MSE: 0.3326)

Época 82/250 [Train] L1=1.95e-03: 100% | 185/185 [00:02<00:00, 80.33it/s, loss=0.3596,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 82 (Val MSE: 0.3324)

Época 83/250 [Train] L1=1.91e-03: 100% | 185/185 [00:02<00:00, 81.80it/s, loss=0.3483,
Época 84/250 [Train] L1=1.87e-03: 100% | 185/185 [00:02<00:00, 80.47it/s, loss=0.3670,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 84 (Val MSE: 0.3320)

Época 85/250 [Train] L1=1.83e-03: 100% | 185/185 [00:02<00:00, 80.86it/s, loss=0.3625,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 85 (Val MSE: 0.3319)

Época 85/250 | L1 coef: 1.83e-03
Train - Total: 0.3530 | MSE: 0.3311 | L0: 66.9
Val - Total: 0.3544 | MSE: 0.3319 | L0: 75.7
Best Val MSE: 0.3319 @ época 85 | Sin mejora: 0/20

Época 86/250 [Train] L1=1.80e-03: 100% | 185/185 [00:02<00:00, 80.21it/s, loss=0.3715,
Época 87/250 [Train] L1=1.76e-03: 100% | 185/185 [00:02<00:00, 81.35it/s, loss=0.3683,
Época 88/250 [Train] L1=1.72e-03: 100% | 185/185 [00:02<00:00, 80.14it/s, loss=0.3448,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 88 (Val MSE: 0.3312)

Época 89/250 [Train] L1=1.69e-03: 100% | 185/185 [00:02<00:00, 80.95it/s, loss=0.3584,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 89 (Val MSE: 0.3312)

Época 90/250 [Train] L1=1.66e-03: 100%| | 185/185 [00:02<00:00, 79.79it/s, loss=0.3554,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 90 (Val MSE: 0.3310)

Época 90/250 | L1 coef: 1.66e-03
Train - Total: 0.3502 | MSE: 0.3301 | L0: 67.6
Val - Total: 0.3516 | MSE: 0.3310 | L0: 76.4
Best Val MSE: 0.3310 @ época 90 | Sin mejora: 0/20

Época 91/250 [Train] L1=1.62e-03: 100%| | 185/185 [00:02<00:00, 81.13it/s, loss=0.3545,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 91 (Val MSE: 0.3307)

Época 92/250 [Train] L1=1.59e-03: 100%| | 185/185 [00:02<00:00, 81.97it/s, loss=0.3394,
Época 93/250 [Train] L1=1.56e-03: 100%| | 185/185 [00:02<00:00, 79.97it/s, loss=0.3520,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 93 (Val MSE: 0.3304)

Época 94/250 [Train] L1=1.53e-03: 100%| | 185/185 [00:02<00:00, 81.87it/s, loss=0.3562,
Época 95/250 [Train] L1=1.50e-03: 100%| | 185/185 [00:02<00:00, 79.78it/s, loss=0.3537,

Época 95/250 | L1 coef: 1.50e-03
Train - Total: 0.3479 | MSE: 0.3296 | L0: 68.1
Val - Total: 0.3498 | MSE: 0.3310 | L0: 76.6
Best Val MSE: 0.3304 @ época 93 | Sin mejora: 2/20

Época 96/250 [Train] L1=1.47e-03: 100%| | 185/185 [00:02<00:00, 81.02it/s, loss=0.3571,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 96 (Val MSE: 0.3300)

Época 97/250 [Train] L1=1.44e-03: 100% | 185/185 [00:02<00:00, 81.02it/s, loss=0.3619,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 97 (Val MSE: 0.3299)

Época 98/250 [Train] L1=1.41e-03: 100% | 185/185 [00:02<00:00, 79.36it/s, loss=0.3445,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 98 (Val MSE: 0.3296)

Época 99/250 [Train] L1=1.38e-03: 100% | 185/185 [00:02<00:00, 77.97it/s, loss=0.3568,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 99 (Val MSE: 0.3296)

Época 100/250 [Train] L1=1.35e-03: 100% | 185/185 [00:02<00:00, 81.91it/s, loss=0.3460,

Época 100/250 | L1 coef: 1.35e-03
Train - Total: 0.3456 | MSE: 0.3288 | L0: 68.7
Val - Total: 0.3470 | MSE: 0.3298 | L0: 77.4
Best Val MSE: 0.3296 @ época 99 | Sin mejora: 1/20

Época 101/250 [Train] L1=1.33e-03: 100% | 185/185 [00:02<00:00, 81.31it/s, loss=0.3586,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 101 (Val MSE: 0.3295)

Época 102/250 [Train] L1=1.30e-03: 100% | 185/185 [00:02<00:00, 80.47it/s, loss=0.3387,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 102 (Val MSE: 0.3294)

Época 103/250 [Train] L1=1.27e-03: 100% | 185/185 [00:02<00:00, 81.45it/s, loss=0.3324,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 103 (Val MSE: 0.3292)

Época 104/250 [Train] L1=1.25e-03: 100% | 185/185 [00:02<00:00, 80.09it/s, loss=0.3445,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 104 (Val MSE: 0.3290)

Época 105/250 [Train] L1=1.22e-03: 100% | 185/185 [00:02<00:00, 80.86it/s, loss=0.3372,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 105 (Val MSE: 0.3288)

Época 105/250 | L1 coef: 1.22e-03
Train - Total: 0.3433 | MSE: 0.3280 | L0: 69.2
Val - Total: 0.3446 | MSE: 0.3288 | L0: 78.0
Best Val MSE: 0.3288 @ época 105 | Sin mejora: 0/20

Época 106/250 [Train] L1=1.20e-03: 100% | 185/185 [00:02<00:00, 77.50it/s, loss=0.3415,
Época 107/250 [Train] L1=1.17e-03: 100% | 185/185 [00:02<00:00, 80.55it/s, loss=0.3582,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 107 (Val MSE: 0.3287)

Época 108/250 [Train] L1=1.15e-03: 100% | 185/185 [00:02<00:00, 79.88it/s, loss=0.3387,
Época 109/250 [Train] L1=1.13e-03: 100% | 185/185 [00:02<00:00, 81.06it/s, loss=0.3453,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 109 (Val MSE: 0.3283)

Época 110/250 [Train] L1=1.11e-03: 100%| | 185/185 [00:02<00:00, 82.16it/s, loss=0.3363,

Época 110/250 | L1 coef: 1.11e-03

Train - Total: 0.3417 | MSE: 0.3276 | L0: 69.6

Val - Total: 0.3429 | MSE: 0.3284 | L0: 78.4

Best Val MSE: 0.3283 @ época 109 | Sin mejora: 1/20

Época 111/250 [Train] L1=1.08e-03: 100%| | 185/185 [00:02<00:00, 77.69it/s, loss=0.3260,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 111 (Val MSE: 0.3282)

Época 112/250 [Train] L1=1.06e-03: 100%| | 185/185 [00:02<00:00, 77.64it/s, loss=0.3268,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 112 (Val MSE: 0.3281)

Época 113/250 [Train] L1=1.04e-03: 100%| | 185/185 [00:02<00:00, 75.59it/s, loss=0.3305,

Época 114/250 [Train] L1=1.02e-03: 100%| | 185/185 [00:02<00:00, 81.53it/s, loss=0.3361,

Época 115/250 [Train] L1=9.99e-04: 100%| | 185/185 [00:02<00:00, 77.19it/s, loss=0.3514,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 115 (Val MSE: 0.3279)

Época 115/250 | L1 coef: 9.99e-04

Train - Total: 0.3401 | MSE: 0.3273 | L0: 70.0

Val - Total: 0.3411 | MSE: 0.3279 | L0: 78.8

Best Val MSE: 0.3279 @ época 115 | Sin mejora: 0/20

Época 116/250 [Train] L1=9.79e-04: 100%| | 185/185 [00:02<00:00, 81.96it/s, loss=0.3502,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 116 (Val MSE: 0.3278)

Época 117/250 [Train] L1=9.60e-04: 100%| | 185/185 [00:02<00:00, 81.59it/s, loss=0.3353,
Época 118/250 [Train] L1=9.41e-04: 100%| | 185/185 [00:02<00:00, 80.21it/s, loss=0.3436,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 118 (Val MSE: 0.3277)

Época 119/250 [Train] L1=9.22e-04: 100%| | 185/185 [00:02<00:00, 80.13it/s, loss=0.3581,
Época 120/250 [Train] L1=9.03e-04: 100%| | 185/185 [00:02<00:00, 80.39it/s, loss=0.3330,

Época 120/250 | L1 coef: 9.03e-04

Train - Total: 0.3387 | MSE: 0.3269 | L0: 70.3

Val - Total: 0.3399 | MSE: 0.3278 | L0: 79.1

Best Val MSE: 0.3277 @ época 118 | Sin mejora: 2/20

Época 121/250 [Train] L1=8.85e-04: 100%| | 185/185 [00:02<00:00, 79.12it/s, loss=0.3256,
Época 122/250 [Train] L1=8.68e-04: 100%| | 185/185 [00:02<00:00, 79.16it/s, loss=0.3493,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 122 (Val MSE: 0.3275)

Época 123/250 [Train] L1=8.50e-04: 100%| | 185/185 [00:02<00:00, 81.79it/s, loss=0.3488,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 123 (Val MSE: 0.3273)

Época 124/250 [Train] L1=8.33e-04: 100%| | 185/185 [00:02<00:00, 80.65it/s, loss=0.3425,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 124 (Val MSE: 0.3272)

Época 125/250 [Train] L1=8.17e-04: 100%| | 185/185 [00:02<00:00, 80.68it/s, loss=0.3485,

Época 125/250 | L1 coef: 8.17e-04
Train - Total: 0.3374 | MSE: 0.3266 | L0: 70.7
Val - Total: 0.3386 | MSE: 0.3275 | L0: 79.4
Best Val MSE: 0.3272 @ época 124 | Sin mejora: 1/20

Época 126/250 [Train] L1=8.00e-04: 100%| | 185/185 [00:02<00:00, 80.36it/s, loss=0.3391,
Época 127/250 [Train] L1=7.84e-04: 100%| | 185/185 [00:02<00:00, 77.49it/s, loss=0.3378,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 127 (Val MSE: 0.3270)

Época 128/250 [Train] L1=7.69e-04: 100%| | 185/185 [00:02<00:00, 78.90it/s, loss=0.3432,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 128 (Val MSE: 0.3270)

Época 129/250 [Train] L1=7.53e-04: 100%| | 185/185 [00:02<00:00, 79.98it/s, loss=0.3488,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 129 (Val MSE: 0.3268)

Época 130/250 [Train] L1=7.38e-04: 100%| | 185/185 [00:02<00:00, 81.17it/s, loss=0.3403,

Época 130/250 | L1 coef: 7.38e-04
Train - Total: 0.3360 | MSE: 0.3261 | L0: 71.0
Val - Total: 0.3371 | MSE: 0.3269 | L0: 79.8
Best Val MSE: 0.3268 @ época 129 | Sin mejora: 1/20

Época 131/250 [Train] L1=7.23e-04: 100%| | 185/185 [00:02<00:00, 81.89it/s, loss=0.3479,
Época 132/250 [Train] L1=7.09e-04: 100%| | 185/185 [00:02<00:00, 78.98it/s, loss=0.3462,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 132 (Val MSE: 0.3268)

Época 133/250 [Train] L1=6.95e-04: 100%| | 185/185 [00:02<00:00, 81.32it/s, loss=0.3305,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 133 (Val MSE: 0.3267)

Época 134/250 [Train] L1=6.81e-04: 100%| | 185/185 [00:02<00:00, 80.84it/s, loss=0.3398,

Época 135/250 [Train] L1=6.67e-04: 100%| | 185/185 [00:02<00:00, 82.02it/s, loss=0.3358,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 135 (Val MSE: 0.3266)

Época 135/250 | L1 coef: 6.67e-04

Train - Total: 0.3349 | MSE: 0.3259 | L0: 71.2

Val - Total: 0.3360 | MSE: 0.3266 | L0: 80.0

Best Val MSE: 0.3266 @ época 135 | Sin mejora: 0/20

Época 136/250 [Train] L1=6.54e-04: 100%| | 185/185 [00:02<00:00, 79.65it/s, loss=0.3334,

Época 137/250 [Train] L1=6.41e-04: 100%| | 185/185 [00:02<00:00, 81.86it/s, loss=0.3230,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 137 (Val MSE: 0.3264)

Época 138/250 [Train] L1=6.28e-04: 100%| | 185/185 [00:02<00:00, 79.46it/s, loss=0.3326,

Época 139/250 [Train] L1=6.15e-04: 100%| | 185/185 [00:02<00:00, 78.21it/s, loss=0.3427,

Época 140/250 [Train] L1=6.03e-04: 100%| | 185/185 [00:02<00:00, 81.01it/s, loss=0.3373,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 140 (Val MSE: 0.3263)

Época 140/250 | L1 coef: 6.03e-04

Train - Total: 0.3339 | MSE: 0.3256 | L0: 71.4

Val - Total: 0.3348 | MSE: 0.3263 | L0: 80.2

Best Val MSE: 0.3263 @ época 140 | Sin mejora: 0/20

Época 141/250 [Train] L1=5.91e-04: 100%| | 185/185 [00:02<00:00, 81.38it/s, loss=0.3300,

Época 142/250 [Train] L1=5.79e-04: 100%| | 185/185 [00:02<00:00, 83.56it/s, loss=0.3311,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 142 (Val MSE: 0.3262)

Época 143/250 [Train] L1=5.68e-04: 100% | 185/185 [00:02<00:00, 90.09it/s, loss=0.3350,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 143 (Val MSE: 0.3262)

Época 144/250 [Train] L1=5.56e-04: 100% | 185/185 [00:02<00:00, 91.08it/s, loss=0.3286,
Época 145/250 [Train] L1=5.45e-04: 100% | 185/185 [00:02<00:00, 90.77it/s, loss=0.3200,

Época 145/250 | L1 coef: 5.45e-04
Train - Total: 0.3329 | MSE: 0.3253 | L0: 71.7
Val - Total: 0.3340 | MSE: 0.3262 | L0: 80.3
Best Val MSE: 0.3262 @ época 143 | Sin mejora: 2/20

Época 146/250 [Train] L1=5.34e-04: 100% | 185/185 [00:02<00:00, 91.73it/s, loss=0.3382,
Época 147/250 [Train] L1=5.24e-04: 100% | 185/185 [00:02<00:00, 90.74it/s, loss=0.3287,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 147 (Val MSE: 0.3261)

Época 148/250 [Train] L1=5.13e-04: 100% | 185/185 [00:02<00:00, 88.02it/s, loss=0.3288,
Época 149/250 [Train] L1=5.03e-04: 100% | 185/185 [00:02<00:00, 92.21it/s, loss=0.3347,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 149 (Val MSE: 0.3260)

Época 150/250 [Train] L1=4.93e-04: 100% | 185/185 [00:02<00:00, 90.93it/s, loss=0.3365,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 150 (Val MSE: 0.3259)

Época 150/250 | L1 coef: 4.93e-04
Train - Total: 0.3321 | MSE: 0.3251 | L0: 71.9
Val - Total: 0.3331 | MSE: 0.3259 | L0: 80.5
Best Val MSE: 0.3259 @ época 150 | Sin mejora: 0/20

Época 151/250 [Train] L1=4.83e-04: 100%| | 185/185 [00:02<00:00, 92.14it/s, loss=0.3324,
Época 152/250 [Train] L1=4.73e-04: 100%| | 185/185 [00:02<00:00, 89.17it/s, loss=0.3020,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 152 (Val MSE: 0.3258)

Época 153/250 [Train] L1=4.64e-04: 100%| | 185/185 [00:02<00:00, 91.05it/s, loss=0.3271,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 153 (Val MSE: 0.3257)

Época 154/250 [Train] L1=4.55e-04: 100%| | 185/185 [00:02<00:00, 90.96it/s, loss=0.3349,
Época 155/250 [Train] L1=4.45e-04: 100%| | 185/185 [00:02<00:00, 89.07it/s, loss=0.3376,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 155 (Val MSE: 0.3257)

Época 155/250 | L1 coef: 4.45e-04
Train - Total: 0.3320 | MSE: 0.3256 | L0: 71.9
Val - Total: 0.3322 | MSE: 0.3257 | L0: 80.7
Best Val MSE: 0.3257 @ época 155 | Sin mejora: 0/20

Época 156/250 [Train] L1=4.37e-04: 100%| | 185/185 [00:02<00:00, 80.82it/s, loss=0.3300,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 156 (Val MSE: 0.3256)

Época 157/250 [Train] L1=4.28e-04: 100%| | 185/185 [00:02<00:00, 81.45it/s, loss=0.3188,
Época 158/250 [Train] L1=4.19e-04: 100%| | 185/185 [00:02<00:00, 78.84it/s, loss=0.3307,
Época 159/250 [Train] L1=4.11e-04: 100%| | 185/185 [00:02<00:00, 80.48it/s, loss=0.3444,
Época 160/250 [Train] L1=4.03e-04: 100%| | 185/185 [00:02<00:00, 79.46it/s, loss=0.3303,

Época 160/250 | L1 coef: 4.03e-04

Train - Total: 0.3316 | MSE: 0.3257 | L0: 72.1

Val - Total: 0.3316 | MSE: 0.3256 | L0: 80.9

Best Val MSE: 0.3256 @ época 156 | Sin mejora: 4/20

Época 161/250 [Train] L1=3.95e-04: 100%| | 185/185 [00:02<00:00, 82.64it/s, loss=0.3250,
Época 162/250 [Train] L1=3.87e-04: 100%| | 185/185 [00:02<00:00, 81.59it/s, loss=0.3317,
Época 163/250 [Train] L1=3.79e-04: 100%| | 185/185 [00:02<00:00, 81.51it/s, loss=0.3133,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 163 (Val MSE: 0.3254)

Época 164/250 [Train] L1=3.71e-04: 100%| | 185/185 [00:02<00:00, 76.67it/s, loss=0.3370,
Época 165/250 [Train] L1=3.64e-04: 100%| | 185/185 [00:02<00:00, 77.80it/s, loss=0.3378,

Época 165/250 | L1 coef: 3.64e-04

Train - Total: 0.3303 | MSE: 0.3249 | L0: 72.3

Val - Total: 0.3309 | MSE: 0.3255 | L0: 81.0

Best Val MSE: 0.3254 @ época 163 | Sin mejora: 2/20

Época 166/250 [Train] L1=3.57e-04: 100%| | 185/185 [00:02<00:00, 79.28it/s, loss=0.3149,
Época 167/250 [Train] L1=3.50e-04: 100%| | 185/185 [00:02<00:00, 76.40it/s, loss=0.3294,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 167 (Val MSE: 0.3253)

Época 168/250 [Train] L1=3.43e-04: 100%| | 185/185 [00:02<00:00, 78.78it/s, loss=0.3241,
Época 169/250 [Train] L1=3.36e-04: 100%| | 185/185 [00:02<00:00, 76.09it/s, loss=0.3343,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 169 (Val MSE: 0.3252)

Época 170/250 [Train] L1=3.29e-04: 100%| | 185/185 [00:02<00:00, 78.56it/s, loss=0.3302,

Época 170/250 | L1 coef: 3.29e-04

Train - Total: 0.3296 | MSE: 0.3247 | L0: 72.5

Val - Total: 0.3305 | MSE: 0.3255 | L0: 81.1

Best Val MSE: 0.3252 @ época 169 | Sin mejora: 1/20

Época 171/250 [Train] L1=3.22e-04: 100%| | 185/185 [00:02<00:00, 77.10it/s, loss=0.3176,

Época 172/250 [Train] L1=3.16e-04: 100%| | 185/185 [00:02<00:00, 77.16it/s, loss=0.3201,

Época 173/250 [Train] L1=3.10e-04: 100%| | 185/185 [00:02<00:00, 75.42it/s, loss=0.3325,

Época 174/250 [Train] L1=3.03e-04: 100%| | 185/185 [00:02<00:00, 76.15it/s, loss=0.3443,

Época 175/250 [Train] L1=2.97e-04: 100%| | 185/185 [00:02<00:00, 76.98it/s, loss=0.3313,

Época 175/250 | L1 coef: 2.97e-04

Train - Total: 0.3291 | MSE: 0.3246 | L0: 72.6

Val - Total: 0.3299 | MSE: 0.3253 | L0: 81.3

Best Val MSE: 0.3252 @ época 169 | Sin mejora: 6/20

Época 176/250 [Train] L1=2.91e-04: 100%| | 185/185 [00:02<00:00, 79.96it/s, loss=0.3223,

Época 177/250 [Train] L1=2.86e-04: 100%| | 185/185 [00:02<00:00, 77.89it/s, loss=0.3228,

Época 178/250 [Train] L1=2.80e-04: 100%| | 185/185 [00:02<00:00, 80.38it/s, loss=0.3230,

Época 179/250 [Train] L1=2.74e-04: 100%| | 185/185 [00:02<00:00, 76.89it/s, loss=0.3176,

Época 180/250 [Train] L1=2.69e-04: 100%| | 185/185 [00:02<00:00, 79.16it/s, loss=0.3294,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 180 (Val MSE: 0.3251)

Época 180/250 | L1 coef: 2.69e-04

Train - Total: 0.3287 | MSE: 0.3246 | L0: 72.6

Val - Total: 0.3293 | MSE: 0.3251 | L0: 81.4

Best Val MSE: 0.3251 @ época 180 | Sin mejora: 0/20

Época 181/250 [Train] L1=2.63e-04: 100%| | 185/185 [00:02<00:00, 82.61it/s, loss=0.3340,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 181 (Val MSE: 0.3250)

Época 182/250 [Train] L1=2.58e-04: 100%| | 185/185 [00:02<00:00, 82.90it/s, loss=0.3270,
Época 183/250 [Train] L1=2.53e-04: 100%| | 185/185 [00:02<00:00, 81.59it/s, loss=0.3288,
Época 184/250 [Train] L1=2.48e-04: 100%| | 185/185 [00:02<00:00, 80.55it/s, loss=0.3234,
Época 185/250 [Train] L1=2.43e-04: 100%| | 185/185 [00:02<00:00, 80.21it/s, loss=0.3261,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 185 (Val MSE: 0.3249)

Época 185/250 | L1 coef: 2.43e-04
Train - Total: 0.3280 | MSE: 0.3243 | L0: 72.8
Val - Total: 0.3287 | MSE: 0.3249 | L0: 81.6
Best Val MSE: 0.3249 @ época 185 | Sin mejora: 0/20

Época 186/250 [Train] L1=2.38e-04: 100%| | 185/185 [00:02<00:00, 81.52it/s, loss=0.3275,
Época 187/250 [Train] L1=2.33e-04: 100%| | 185/185 [00:02<00:00, 79.26it/s, loss=0.3132,
Época 188/250 [Train] L1=2.29e-04: 100%| | 185/185 [00:02<00:00, 80.87it/s, loss=0.3309,
Época 189/250 [Train] L1=2.24e-04: 100%| | 185/185 [00:02<00:00, 78.25it/s, loss=0.3322,
Época 190/250 [Train] L1=2.20e-04: 100%| | 185/185 [00:02<00:00, 78.09it/s, loss=0.3255,

Época 190/250 | L1 coef: 2.20e-04
Train - Total: 0.3279 | MSE: 0.3244 | L0: 72.8
Val - Total: 0.3289 | MSE: 0.3254 | L0: 81.5
Best Val MSE: 0.3249 @ época 185 | Sin mejora: 5/20

Época 191/250 [Train] L1=2.15e-04: 100%| | 185/185 [00:02<00:00, 79.22it/s, loss=0.3241,
Época 192/250 [Train] L1=2.11e-04: 100%| | 185/185 [00:02<00:00, 79.46it/s, loss=0.3531,
Época 193/250 [Train] L1=2.07e-04: 100%| | 185/185 [00:02<00:00, 79.28it/s, loss=0.3183,
Época 194/250 [Train] L1=2.03e-04: 100%| | 185/185 [00:02<00:00, 77.26it/s, loss=0.3237,
Época 195/250 [Train] L1=1.99e-04: 100%| | 185/185 [00:02<00:00, 79.28it/s, loss=0.3160,

Época 195/250 | L1 coef: 1.99e-04
Train - Total: 0.3274 | MSE: 0.3243 | L0: 72.9
Val - Total: 0.3285 | MSE: 0.3252 | L0: 81.6
Best Val MSE: 0.3249 @ época 185 | Sin mejora: 10/20

Época 196/250 [Train] L1=1.95e-04: 100%| | 185/185 [00:02<00:00, 78.31it/s, loss=0.3458,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 196 (Val MSE: 0.3249)

Época 197/250 [Train] L1=1.91e-04: 100%| | 185/185 [00:02<00:00, 80.51it/s, loss=0.3268,
Época 198/250 [Train] L1=1.87e-04: 100%| | 185/185 [00:02<00:00, 80.73it/s, loss=0.3243,
Época 199/250 [Train] L1=1.83e-04: 100%| | 185/185 [00:02<00:00, 77.84it/s, loss=0.3336,
Época 200/250 [Train] L1=1.79e-04: 100%| | 185/185 [00:02<00:00, 79.08it/s, loss=0.3238,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 200 (Val MSE: 0.3248)

Época 200/250 | L1 coef: 1.79e-04

Train - Total: 0.3269 | MSE: 0.3241 | L0: 73.0

Val - Total: 0.3277 | MSE: 0.3248 | L0: 81.7

Best Val MSE: 0.3248 @ época 200 | Sin mejora: 0/20

Época 201/250 [Train] L1=1.76e-04: 100%| | 185/185 [00:02<00:00, 82.37it/s, loss=0.3303,
Época 202/250 [Train] L1=1.72e-04: 100%| | 185/185 [00:02<00:00, 79.14it/s, loss=0.3116,
Época 203/250 [Train] L1=1.69e-04: 100%| | 185/185 [00:02<00:00, 78.36it/s, loss=0.3205,
Época 204/250 [Train] L1=1.66e-04: 100%| | 185/185 [00:02<00:00, 80.06it/s, loss=0.3273,
Época 205/250 [Train] L1=1.62e-04: 100%| | 185/185 [00:02<00:00, 77.60it/s, loss=0.3352,

Época 205/250 | L1 coef: 1.62e-04

Train - Total: 0.3275 | MSE: 0.3249 | L0: 73.0

Val - Total: 0.3275 | MSE: 0.3248 | L0: 81.8

Best Val MSE: 0.3248 @ época 200 | Sin mejora: 5/20

Época 206/250 [Train] L1=1.59e-04: 100%| | 185/185 [00:02<00:00, 81.98it/s, loss=0.3250,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 206 (Val MSE: 0.3247)

Época 207/250 [Train] L1=1.56e-04: 100%| | 185/185 [00:02<00:00, 80.72it/s, loss=0.3209,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 207 (Val MSE: 0.3246)

Época 208/250 [Train] L1=1.53e-04: 100%| | 185/185 [00:02<00:00, 79.38it/s, loss=0.3373,
Época 209/250 [Train] L1=1.50e-04: 100%| | 185/185 [00:02<00:00, 81.03it/s, loss=0.3185,
Época 210/250 [Train] L1=1.47e-04: 100%| | 185/185 [00:02<00:00, 79.82it/s, loss=0.3253,

Época 210/250 | L1 coef: 1.47e-04
Train - Total: 0.3262 | MSE: 0.3238 | L0: 73.2
Val - Total: 0.3271 | MSE: 0.3246 | L0: 81.9
Best Val MSE: 0.3246 @ época 207 | Sin mejora: 3/20

Época 211/250 [Train] L1=1.44e-04: 100%| | 185/185 [00:02<00:00, 80.49it/s, loss=0.3382,
Época 212/250 [Train] L1=1.41e-04: 100%| | 185/185 [00:02<00:00, 80.22it/s, loss=0.3173,
Época 213/250 [Train] L1=1.38e-04: 100%| | 185/185 [00:02<00:00, 81.02it/s, loss=0.3257,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 213 (Val MSE: 0.3246)

Época 214/250 [Train] L1=1.35e-04: 100%| | 185/185 [00:02<00:00, 80.24it/s, loss=0.3382,
Época 215/250 [Train] L1=1.33e-04: 100%| | 185/185 [00:02<00:00, 80.67it/s, loss=0.3255,

Época 215/250 | L1 coef: 1.33e-04
Train - Total: 0.3262 | MSE: 0.3240 | L0: 73.2
Val - Total: 0.3276 | MSE: 0.3253 | L0: 81.9
Best Val MSE: 0.3246 @ época 213 | Sin mejora: 2/20

Época 216/250 [Train] L1=1.30e-04: 100%| | 185/185 [00:02<00:00, 81.20it/s, loss=0.3176,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 216 (Val MSE: 0.3245)

Época 217/250 [Train] L1=1.27e-04: 100%| | 185/185 [00:02<00:00, 82.14it/s, loss=0.3307,
Época 218/250 [Train] L1=1.25e-04: 100%| | 185/185 [00:02<00:00, 79.69it/s, loss=0.3349,
Época 219/250 [Train] L1=1.22e-04: 100%| | 185/185 [00:02<00:00, 80.47it/s, loss=0.3250,
Época 220/250 [Train] L1=1.20e-04: 100%| | 185/185 [00:02<00:00, 80.53it/s, loss=0.3231,

Época 220/250 | L1 coef: 1.20e-04
Train - Total: 0.3260 | MSE: 0.3240 | L0: 73.2
Val - Total: 0.3266 | MSE: 0.3246 | L0: 82.1
Best Val MSE: 0.3245 @ época 216 | Sin mejora: 4/20

Época 221/250 [Train] L1=1.17e-04: 100%| | 185/185 [00:02<00:00, 82.43it/s, loss=0.3285,
Época 222/250 [Train] L1=1.15e-04: 100%| | 185/185 [00:02<00:00, 79.97it/s, loss=0.3380,
Época 223/250 [Train] L1=1.13e-04: 100%| | 185/185 [00:02<00:00, 78.10it/s, loss=0.3226,
Época 224/250 [Train] L1=1.11e-04: 100%| | 185/185 [00:02<00:00, 81.81it/s, loss=0.3259,
Época 225/250 [Train] L1=1.08e-04: 100%| | 185/185 [00:02<00:00, 81.08it/s, loss=0.3230,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 225 (Val MSE: 0.3245)

Época 225/250 | L1 coef: 1.08e-04
Train - Total: 0.3259 | MSE: 0.3241 | L0: 73.3
Val - Total: 0.3264 | MSE: 0.3245 | L0: 82.0
Best Val MSE: 0.3245 @ época 225 | Sin mejora: 0/20

Época 226/250 [Train] L1=1.06e-04: 100%| | 185/185 [00:02<00:00, 81.55it/s, loss=0.3373,
Época 227/250 [Train] L1=1.04e-04: 100%| | 185/185 [00:02<00:00, 80.84it/s, loss=0.3151,
Época 228/250 [Train] L1=1.02e-04: 100%| | 185/185 [00:02<00:00, 78.86it/s, loss=0.3470,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 228 (Val MSE: 0.3245)

Época 229/250 [Train] L1=9.99e-05: 100%| | 185/185 [00:02<00:00, 80.16it/s, loss=0.3213,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt
Mejor modelo guardado en época 229 (Val MSE: 0.3244)

Época 230/250 [Train] L1=9.79e-05: 100%| | 185/185 [00:02<00:00, 78.93it/s, loss=0.3182,

Época 230/250 | L1 coef: 9.79e-05
Train - Total: 0.3253 | MSE: 0.3237 | L0: 73.4
Val - Total: 0.3264 | MSE: 0.3247 | L0: 82.3
Best Val MSE: 0.3244 @ época 229 | Sin mejora: 1/20

Época 231/250 [Train] L1=9.59e-05: 100%| | 185/185 [00:02<00:00, 79.08it/s, loss=0.3331,
Época 232/250 [Train] L1=9.40e-05: 100%| | 185/185 [00:02<00:00, 77.03it/s, loss=0.3164,
Época 233/250 [Train] L1=9.21e-05: 100%| | 185/185 [00:02<00:00, 80.87it/s, loss=0.3365,
Época 234/250 [Train] L1=9.03e-05: 100%| | 185/185 [00:02<00:00, 79.75it/s, loss=0.3357,
Época 235/250 [Train] L1=8.85e-05: 100%| | 185/185 [00:02<00:00, 80.11it/s, loss=0.3313

Época 235/250 | L1 coef: 8.85e-05

Train - Total: 0.3255 | MSE: 0.3240 | L0: 73.4

Val - Total: 0.3260 | MSE: 0.3245 | L0: 82.2

Best Val MSE: 0.3244 @ época 229 | Sin mejora: 6/20

Época 236/250 [Train] L1=8.67e-05: 100%| | 185/185 [00:02<00:00, 78.82it/s, loss=0.3220,
Época 237/250 [Train] L1=8.50e-05: 100%| | 185/185 [00:02<00:00, 78.00it/s, loss=0.3182,
Época 238/250 [Train] L1=8.33e-05: 100%| | 185/185 [00:02<00:00, 80.92it/s, loss=0.3164,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_best.pt

Mejor modelo guardado en época 238 (Val MSE: 0.3243)

Época 239/250 [Train] L1=8.16e-05: 100%| | 185/185 [00:02<00:00, 81.10it/s, loss=0.3249,
Época 240/250 [Train] L1=8.00e-05: 100%| | 185/185 [00:02<00:00, 81.11it/s, loss=0.3290,

Época 240/250 | L1 coef: 8.00e-05

Train - Total: 0.3255 | MSE: 0.3241 | L0: 73.4

Val - Total: 0.3263 | MSE: 0.3249 | L0: 82.1

Best Val MSE: 0.3243 @ época 238 | Sin mejora: 2/20

Época 241/250 [Train] L1=7.84e-05: 100%| | 185/185 [00:02<00:00, 82.04it/s, loss=0.3243,
Época 242/250 [Train] L1=7.68e-05: 100%| | 185/185 [00:02<00:00, 81.79it/s, loss=0.3257,
Época 243/250 [Train] L1=7.53e-05: 100%| | 185/185 [00:02<00:00, 80.98it/s, loss=0.3236,
Época 244/250 [Train] L1=7.38e-05: 100%| | 185/185 [00:02<00:00, 78.48it/s, loss=0.3306,
Época 245/250 [Train] L1=7.23e-05: 100%| | 185/185 [00:02<00:00, 79.58it/s, loss=0.3318,

Época 245/250 | L1 coef: 7.23e-05

Train - Total: 0.3253 | MSE: 0.3240 | L0: 73.4

Val - Total: 0.3263 | MSE: 0.3250 | L0: 82.2

Best Val MSE: 0.3243 @ época 238 | Sin mejora: 7/20

```

Época 246/250 [Train] L1=7.09e-05: 100%|      | 185/185 [00:02<00:00, 80.53it/s, loss=0.3215,
Época 247/250 [Train] L1=6.94e-05: 100%|      | 185/185 [00:02<00:00, 80.23it/s, loss=0.3279,
Época 248/250 [Train] L1=6.81e-05: 100%|      | 185/185 [00:02<00:00, 80.50it/s, loss=0.3277,
Época 249/250 [Train] L1=6.67e-05: 100%|      | 185/185 [00:02<00:00, 81.32it/s, loss=0.3095,
Época 250/250 [Train] L1=6.54e-05: 100%|      | 185/185 [00:02<00:00, 81.52it/s, loss=0.3369

```

```

Época 250/250 | L1 coef: 6.54e-05
Train - Total: 0.3250 | MSE: 0.3239 | L0: 73.5
Val   - Total: 0.3257 | MSE: 0.3245 | L0: 82.2
Best Val MSE: 0.3243 @ época 238 | Sin mejora: 12/20

```

```

=====
RESUMEN DEL ENTRENAMIENTO
=====
Mejor modelo en época 238 con Val MSE = 0.3243
Early stopping tras 250 épocas

Métricas Finales (Última Época):
      Métrica  Train Validation
Total Loss 0.3250      0.3257
MSE Loss 0.3239      0.3245
L0 Sparsity 73.5      82.2
Fraction Active 0.45%    0.50%
=====

```

8. Guardar Modelo

```

# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='final'
    )
    checkpoint_path = CONFIG['save_dir'] / final_model_name
    torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex4_final.pt

```

=====
CHECKPOINTS GUARDADOS

```

```

=====
Tipo                                Ubicación
Local saved_model\sae_standard_l1-decay_l6_1000g_final.pt
Organizado                          layer_06/models/...
=====

```

9. Métricas básicas del modelo

```

# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

```

```

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print("" + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

```

=====
METRICAS DE ENTRENAMIENTO GUARDADAS
=====

```

Metrica	Valor
Val MSE (final)	0.324548
Val MSE (mejor)	0.324309
Val L0 (final)	82.2
Fraccion Activa	0.50%

```

Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard
decay_standard\1000games\ex4\metrics\training_metrics.json
=====

```

10. Visualización de Resultados

```

# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')

```

```

axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap

```

```

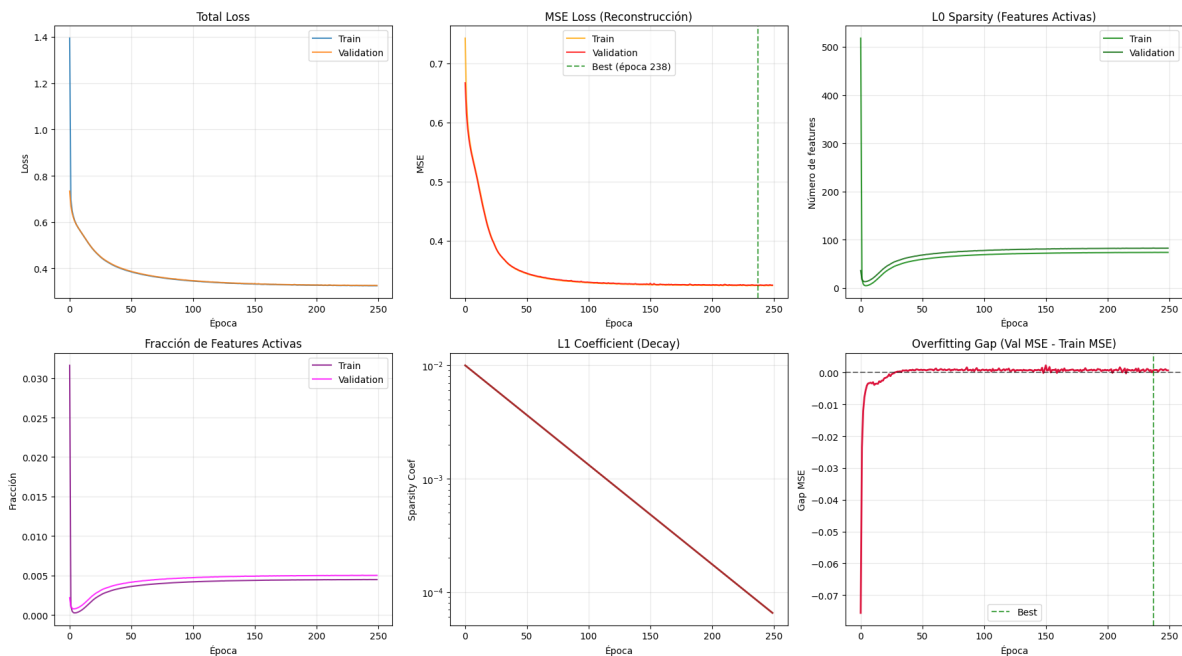
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best Epoch')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')

print('='*50)

```



=====

Gráficas guardadas exitosamente

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):}",
        f"{len(val_dataset):}",
        f"{len(full_dataset):}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",

```



```

        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====

```

Arquitectura:

Componente	Valor
Input	512
Hidden	16384 (32.0x)
Parámetros	16,794,112

Datos:

Conjunto	Muestras
Train	47,200
Validation	11,800
Total	59,000

Métricas Finales:

Métrica	Valor
---------	-------

```
Train Loss 0.3250
  Val Loss 0.3257
Train MSE 0.3239
  Val MSE 0.3245
Train L0    73.5
  Val L0    82.2
```

Archivos Guardados:

```
  Tipo
  Modelo
```

decay_l6_1000g_final.pt

Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard

decay_standard\1000games\ex4\plots\training_curves.png

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
```

```

)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

```

Generando PDF con Quarto...
Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard_l1-decay_16_1000g_ex4_training.pdf
PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard_l1-decay_16_1000g_ex4_training.pdf

```